Improper Integral with Conditional Convergence

Topic.

Improper integrals, absolute convergence, conditional convergence, Dirichlet's test.

Statement.

Investigate absolute and conditional convergence for all

$$\alpha \in \mathbb{R}$$

of the improper integral

$$\int_0^{+\infty} \frac{x^5 \sin(x^2)}{(1+x^4)^\alpha}\, dx.$$

Solution.

First consider the convergence of

$$\int_0^{+\infty} \frac{x^5 \sin(x^2)}{(1+x^4)^\alpha}\, dx.$$

Make the substitution

$$t = x^2, \qquad dt = 2x\, dx.$$

Then

$$x^5\, dx = x^4 \cdot x\, dx = t^2 \cdot \frac{dt}{2}.$$

Therefore,

$$\int_0^{+\infty} \frac{x^5 \sin(x^2)}{(1+x^4)^\alpha}\, dx = \frac{1}{2} \int_0^{+\infty} \frac{t^2 \sin t}{(1+t^2)^\alpha}\, dt.$$

Let

$$g(t) = \frac{t^2}{(1+t^2)^\alpha}, \qquad f(t) = \sin t.$$

The antiderivative of $\sin t$ is bounded:

$$\int \sin t\, dt = -\cos t.$$

Now consider the behavior of $g(t)$ as $t \to +\infty$.

Since

$$1 + t^2 \sim t^2,$$

we get

$$g(t) = \frac{t^2}{(1+t^2)^\alpha} \sim \frac{t^2}{t^{2\alpha}} = t^{2-2\alpha}.$$

By Dirichlet's test, we need $g(t)$ to decrease to zero. Therefore,

$$g(t) \to 0$$

is equivalent to

$$2 - 2\alpha < 0.$$

Hence,

$$\alpha > 1.$$

Thus, for

$$\alpha > 1$$

the integral converges.

Now consider the case

$$\alpha \leq 1.$$

If $\alpha < 1$, then

$$g(t) \sim t^{2-2\alpha} \to +\infty.$$

If $\alpha = 1$, then

$$g(t) = \frac{t^2}{1+t^2} \to 1.$$

Therefore, for $\alpha \leq 1$ there exist constants $C > 0$ and $N > 0$ such that for all $t \geq N$,

$$g(t) \geq C.$$

Take the intervals

$$a_n = 2\pi n + \frac{\pi}{6}, \qquad b_n = 2\pi n + \frac{5\pi}{6}.$$

On each interval $[a_n, b_n]$ we have

$$\sin t \geq \frac{1}{2}.$$

For large n,

$$\int_{a_n}^{b_n} g(t) \sin t \, dt \geq \frac{C}{2}(b_n - a_n).$$

Since

$$b_n - a_n = \frac{2\pi}{3},$$

we get

$$\int_{a_n}^{b_n} g(t) \sin t \, dt \geq \frac{C}{2} \cdot \frac{2\pi}{3} = \frac{C\pi}{3} > 0.$$

Therefore, by Cauchy's criterion, the integral diverges for

$$\alpha \leq 1.$$

Now investigate absolute convergence.

We need to study

$$\int_0^{+\infty} \left| \frac{x^5 \sin(x^2)}{(1+x^4)^\alpha} \right| dx.$$

After the same substitution $t = x^2$, this becomes

$$\frac{1}{2} \int_0^{+\infty} \frac{t^2 |\sin t|}{(1+t^2)^\alpha} \, dt.$$

Split it into two parts:

$$\frac{1}{2} \int_0^1 \frac{t^2 |\sin t|}{(1+t^2)^\alpha} \, dt + \frac{1}{2} \int_1^{+\infty} \frac{t^2 |\sin t|}{(1+t^2)^\alpha} \, dt.$$

The first integral is proper, so it converges.

Now consider

$$\int_1^{+\infty} \frac{t^2|\sin t|}{(1+t^2)^\alpha}\,dt.$$

As $t \to +\infty$,

$$\frac{t^2|\sin t|}{(1+t^2)^\alpha} \sim t^{2-2\alpha}|\sin t|.$$

Since

$$0 \leq |\sin t| \leq 1,$$

we have

$$0 \leq t^{2-2\alpha}|\sin t| \leq t^{2-2\alpha}.$$

By the comparison test, if

$$\int_1^{+\infty} t^{2-2\alpha}\,dt$$

converges, then the integral with $|\sin t|$ also converges.

The integral

$$\int_1^{+\infty} t^{2-2\alpha}\,dt$$

converges when

$$2 - 2\alpha < -1.$$

This is equivalent to

$$\alpha > \frac{3}{2}.$$

Therefore, the original integral converges absolutely for

$$\alpha > \frac{3}{2}.$$

Now let

$$\alpha \le \frac{3}{2}.$$

Again,

$$g(t) = \frac{t^2}{(1+t^2)^\alpha} \sim t^{2-2\alpha}.$$

Therefore, for sufficiently large t there exist constants $C > 0$ and $N > 0$ such that for all $t \ge N$,

$$g(t) \ge Ct^{2-2\alpha}.$$

Take again

$$a_n = 2\pi n + \frac{\pi}{6}, \qquad b_n = 2\pi n + \frac{5\pi}{6}.$$

On $[a_n, b_n]$,

$$|\sin t| \ge \frac{1}{2}.$$

For large n,

$$\int_{a_n}^{b_n} g(t)|\sin t|\,dt \ge \frac{C}{2}\int_{a_n}^{b_n} t^{2-2\alpha}\,dt.$$

The length of the interval is constant, and on this interval t is of order n. Hence

$$\int_{a_n}^{b_n} t^{2-2\alpha}\,dt \ge C_1 n^{2-2\alpha}$$

for some constant $C_1 > 0$.

If

$$1 < \alpha \le \frac{3}{2},$$

then

$$2 - 2\alpha \ge -1.$$

Thus these pieces do not give absolute convergence, and the corresponding series of lower estimates diverges.

Therefore, the integral is not absolutely convergent for

$$1 < \alpha \leq \frac{3}{2}.$$

Combining the results:

$$\boxed{\text{the integral converges absolutely for } \alpha > \frac{3}{2};}$$

$$\boxed{\text{the integral converges conditionally for } 1 < \alpha \leq \frac{3}{2};}$$

$$\boxed{\text{the integral diverges for } \alpha \leq 1.}$$